MATH 110 — Week 5

Notes Chapters 9–10; Homework 5

§1 Quotient maps and the first isomorphism theorem

The question. Which information about a vector is determined by its image under a linear map? For $T(x, y) = x$, inputs with the same first coordinate have the same image; their differences lie in $\ker T = \mathbb{R}(0, 1)$.

Definition (Notes Definition 25; LADR 3.104)**.** For a subspace $U \subseteq V$, the quotient map is $\pi : V \to V/U$, $\pi(v) = v+U$. Cosets form a vector space: $(v + U) + (w + U) = (v + w) + U$ and $a(v + U) = av + U$, with zero coset U. Changing representatives changes either result by an element of U, so these operations are well-defined. Also $v + U = w + U$ iff $v - w \in U$: membership of v in $w + U$ proves one direction, and $v + u = w + (v - w + u)$ gives one inclusion; exchanging v, w gives equality (Week 4 Proposition 39).

Earlier tools used below

1. For a finite-dimensional domain, $\dim V = \dim \ker T + \dim \operatorname{range} T$ (Week 3 Proposition 24). Extending a kernel basis by v_j gives a range basis Tv_j: a relation among these images puts the same combination of the v_j in the kernel, forcing every coefficient to zero; expanding a domain vector proves spanning.
2. A linear bijection is an isomorphism (Week 4 Proposition 29): if S is its set inverse, both $S(aw_1 + w_2)$ and $aSw_1 + Sw_2$ have image $aw_1 + w_2$, so injectivity proves S linear. Between equal finite dimensions, zero kernel forces full range by the preceding dimension equation and the full-dimension subspace criterion (Week 2 Proposition 14).

Proposition (Notes Definition 25; LADR 3.104)

The quotient map is linear and surjective, with kernel U.

Proof. 1. $\pi(av + w) = (av + w) + U = a(v + U) + (w + U) = a\pi(v) + \pi(w)$. Every coset is $\pi(v)$ for some v, giving surjectivity.
2. $v \in \ker \pi$ iff $v + U = U$, which by Week 4 Proposition 39 is equivalent to $v \in U$.

□

Proposition (Notes Proposition 40; LADR 3.105)

If V is finite-dimensional, $\dim(V/U) = \dim V - \dim U$.

Proof. 1. Week 3 rank–nullity gives $\dim V = \dim \ker \pi + \dim \operatorname{range} \pi$.
2. Substitute $\ker \pi = U$ and $\operatorname{range} \pi = V/U$, then rearrange. The range is finite-dimensional because it is the image of a finite-dimensional domain.

□

Construct a quotient basis

1. Extend a basis $u_1, \dots, u_k$ of U to a basis $u_1, \dots, u_k, v_1, \dots, v_r$ of V.
2. The cosets $v_1 + U, \dots, v_r + U$ span: in a basis expansion, the u_i terms belong to the zero coset.
3. If $\sum a_j(v_j + U) = U$, then $\sum a_j v_j \in U$, producing a relation in the full basis; all $a_j = 0$. Thus the cosets form a basis.
4. For $U = \operatorname{span}(e_1, e_2) \subseteq \mathbb{R}^3$, the single coset $e_3 + U$ is a quotient basis.

Proposition (Notes Proposition 42)

If $T : V \to W$ is linear, $U_1 \subseteq V$ and $U_2 \subseteq W$ are subspaces, and $T(U_1) \subseteq U_2$, then $\overline{T} : V/U_1 \to W/U_2$, $\overline{T}(v + U_1) = Tv + U_2$, is well-defined and linear. This condition is also necessary for that rule to be well-defined.

Where the proof comes from

Want. Ensure $v + U_1 \mapsto Tv + U_2$ is a function on cosets.

Key step. Changing v by $u \in U_1$ changes Tv by Tu. That leaves the output coset unchanged exactly when $Tu \in U_2$ for every such u. This explains the condition $T(U_1) \subseteq U_2$.

Proof. 1. If $v + U_1 = w + U_1$, then $v - w \in U_1$, hence $Tv - Tw \in U_2$. Thus $Tv + U_2 = Tw + U_2$.
2. On representative combinations, $\overline{T}((av + w) + U_1) = T(av + w) + U_2 = a(Tv + U_2) + (Tw + U_2)$, proving linearity.
3. Conversely, if the rule is well-defined, for $u \in U_1$ the equality $u + U_1 = 0 + U_1$ forces $Tu + U_2 = U_2$, hence $Tu \in U_2$.

□

Example (An induced map with nonzero outputs)

For $T(x, y, z) = (2x, 3y, 4z)$ on $\mathbb{R}^3$ and $U = \mathbb{R}e_1$, changing (x, y, z) to $(x + t, y, z)$ changes the output by $(2t, 0, 0) \in U$. Identifying V/U with the coordinates (y, z) therefore gives the induced rule $(y, z) \mapsto (3y, 4z)$.

Proposition (Factorization through a quotient; HW5 P2)

For linear $T : V \to W$ and the quotient map $\pi : V \to V/U$, there is a unique linear $\widetilde{T} : V/U \to W$ with $T = \widetilde{T}\pi$ iff $U \subseteq \ker T$. Its rule is $\widetilde{T}(v + U) = Tv$, and

$$\ker \widetilde{T} = (\ker T)/U, \qquad \operatorname{range} \widetilde{T} = \operatorname{range} T.$$

Proof. 1. If $T = \widetilde{T}\pi$, then $u \in U$ gives $Tu = \widetilde{T}(U) = 0$, proving necessity. Conversely $v - w \in U \subseteq \ker T$ gives $Tv = Tw$, the same representative test as Proposition 42 with target subspace zero.
2. The rule is linear since $T(av + w) = aTv + Tw$. Every coset equals $\pi(v)$, so $T = \widetilde{T}\pi$ forces this rule and proves uniqueness.
3. $\widetilde{T}(v + U) = 0$ iff $v \in \ker T$, giving exactly the displayed kernel cosets. The outputs are precisely the vectors Tv, giving the range formula.

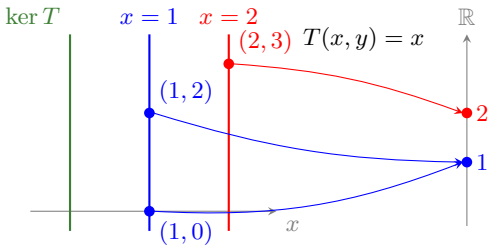
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Proposition (Notes Proposition 41; LADR 3.106–3.107)

For $T : V \to W$ linear, the rule $\overline{T} : V/\ker T \to \operatorname{range} T$, $\overline{T}(v + \ker T) = Tv$, is an isomorphism. If $\iota : \operatorname{range} T \to W$ is the inclusion $w \mapsto w$, then $T = \iota\overline{T}\pi$.

Proof. 1. Apply the factorization criterion with $U = \ker T$. It gives a linear map whose kernel is $(\ker T)/(\ker T) = \{\ker T\}$, the zero subspace of the quotient. Hence it is injective.
2. Its range is $\operatorname{range} T$; choosing this as the codomain makes it onto, hence an isomorphism. For every v, $\iota\overline{T}\pi(v) = Tv$, proving the factorization with inclusion.

□



The two blue inputs differ by $(0, 2) \in \ker T$ and both map to 1. Each vertical coset has one common output. The induced map sends the blue coset to 1 and the distinct red coset to 2.

Example (Notes Example 43)

Differentiation $D : \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$ is onto and has kernel the constants $\mathbb{R}$, as proved in Week 4.

1. Proposition 41 gives $\overline{D} : \mathcal{P}(\mathbb{R})/\mathbb{R} \to \mathcal{P}(\mathbb{R})$, $p + \mathbb{R} \mapsto p'$.
2. Its inverse sends q to the coset $r + \mathbb{R}$ of any antiderivative r of q. Such an r exists by termwise integration; any two differ by a constant, so the coset is unique.
3. For $q = x^2$, the inverse image is $\frac{1}{3}x^3 + \mathbb{R}$. Thus the indefinite integral denotes one quotient vector rather than one chosen polynomial.

Maps and subspaces.

1. An injective $T : V \to W$ gives an isomorphism $V \to \operatorname{range} T$ by restricting its codomain; it is onto that range by definition.
2. A surjective $T : V \to W$ gives $V/\ker T \cong W$ by Proposition 41.
3. In finite dimension, $\dim(V/\ker T) = \dim V - \dim \ker T = \dim \operatorname{range} T$. This recovers the rank–nullity equality, but is not an independent proof because Proposition 40 used rank–nullity.

Shortcuts

- To find a quotient basis, extend a subspace basis and retain the added cosets: the relation test above proves independence. For $\mathbb{R}^3/\operatorname{span}(e_1, e_2)$ this retains $e_3 + U$.
- To identify a quotient with a concrete space, find a surjective map with the specified subspace as kernel; Proposition 41 supplies the isomorphism.

Traps

- The rule $v + U \mapsto Tv$ need not be well-defined unless $U \subseteq \ker T$: with $T = I$ on $\mathbb{R}^2$ and $U = \mathbb{R}e_1$, the same coset $U = 0 + U = e_1 + U$ would have different images.
- Infinite-dimensional V need not give an infinite-dimensional quotient: evaluation $p \mapsto p(0)$ identifies $\mathcal{P}(\mathbb{R})/x\mathcal{P}(\mathbb{R})$ with $\mathbb{R}$ by Proposition 41. On the other hand, $\mathcal{P}(\mathbb{R})/\{0\} \cong \mathcal{P}(\mathbb{R})$ is infinite-dimensional: the cosets of all monomials remain independent.

§2 Successive quotients

The question. In $\mathbb{R}^3$, first identify vectors differing in the x coordinate, then identify changes in y. Both $(2, 3, 5)$ and $(9, -1, 5)$ then represent the same final element, determined by $z = 5$. How can one quotient give the same identification?

Proposition (Notes Proposition 43: second isomorphism theorem)

For subspaces $U, W \subseteq V$, $\phi : U/(U \cap W) \to (U + W)/W$, $\phi(u + (U \cap W)) = u + W$, is an isomorphism. In finite dimension it gives $\dim U - \dim(U \cap W) = \dim(U + W) - \dim W$.

Proof. 1. Every target is $(u+w)+W = u+W$, so use $R : U \to (U+W)/W$, $Ru = u+W$. It is linear by quotient operations; this choice already reaches every target.
2. Two choices $u, u' \in U$ give the same target iff $u - u' \in W$, equivalently $u - u' \in U \cap W$. Thus the source quotient identifies exactly the inputs with equal output.
3. $Ru = W$ iff $u \in W$; since the domain is U, $\ker R = U \cap W$. Proposition 41 supplies the stated isomorphism.
4. If the spaces are finite-dimensional, apply Proposition 40 to both quotients and use equal dimensions of isomorphic spaces.

□

Meaning of the chosen isomorphism. Here the formula for ϕ is specified directly by the subspaces, without choosing bases or complements. This is the sense in which these isomorphisms are called natural in the notes.

Proposition (Notes Proposition 44: third isomorphism theorem)

For subspaces $W \subseteq U \subseteq V$, U/W is the subspace $\{u + W : u \in U\}$ of V/W, and $\phi : (V/W)/(U/W) \to V/U$, $\phi((v+W)+(U/W)) = v+U$, is an isomorphism.

Proof. 1. The set U/W contains the zero coset and is closed under linear combinations because U is a subspace, so it is a subspace of V/W.
2. Since $I(W) \subseteq U$, Proposition 42 gives the linear map $R : V/W \to V/U$, $R(v+W) = v+U$. It is onto because every $v+U$ has this form.
3. $R(v+W) = U$ iff $v \in U$, so $\ker R = U/W$. Proposition 41 gives the displayed formula and isomorphism.
4. For finite-dimensional V, its source dimension is $(\dim V - \dim W) - (\dim U - \dim W) = \dim V - \dim U$, agreeing with the target.

□

Identify successive quotients in coordinates

1. In $V = \mathbb{R}^3$, take $W = \mathbb{R}e_1$ and $U = \operatorname{span}(e_1, e_2)$. A coset of W is determined by coordinates (y, z): representative differences in W change only x.
2. The subspace U/W consists of pairs $(y, 0)$. A coset modulo it is determined by z.
3. Directly, a coset in V/U is also determined by z, and Proposition 44 identifies the two descriptions by keeping that coordinate.

$$(x,y,z) \xrightarrow{\text{modulo } W = \mathbb{R}e_1} (y,z) \xrightarrow{\text{modulo } U/W} z$$

$(2,3,5)$	$(3,5)$	5
$(9,-1,5)$	$(-1,5)$	5

For $W = \mathbb{R}e_1 \subset U = \operatorname{span}(e_1, e_2)$, the first quotient identifies changes in x, and the second identifies changes in y. Both displayed inputs therefore give the same final coset, determined by $z = 5$, just as in V/U.

Shortcuts

- For an induced map, check $T(U_1) \subseteq U_2$ before doing any calculations.
- To prove a quotient isomorphism, start from a linear map on the original space and identify its kernel and range.

Traps

- Nested quotients require $W \subseteq U$: for $U = \mathbb{R}e_1$ and $W = \mathbb{R}e_2$ in $\mathbb{R}^2$, the set W is not a subspace of U, so the quotient U/W is not defined.
- Equal dimensions do not prove a specified map is invertible: the zero map $\mathbb{R} \to \mathbb{R}$ has equal domain and codomain dimensions but is not an isomorphism.

§3 Linear functionals and dual bases

The question. How can a vector be described by linear scalar values? On F^n, the jth coordinate is a linear functional, and all coordinates together determine the vector.

Definition (Notes Definition 26; LADR 3.108, 3.110)**.** A linear functional on V is a linear map $\varphi : V \to F$. The dual space is $V' = L(V, F)$; its elements are also called linear forms, covectors or one-forms. For functionals φ, ψ, their pointwise combination satisfies $(a\varphi + \psi)(bv + w) = b(a\varphi + \psi)(v) + (a\varphi + \psi)(w)$; the zero map is linear. Thus V' is a vector space (Week 3 Proposition 18). The prime on V' denotes duality, not differentiation of a polynomial.

Example (LADR 3.109: functionals)
1. On F^n, $\varphi(x_1, \dots, x_n) = \sum_j a_j x_j$ is linear by distributivity, for any fixed scalars a_j. For example, on $\mathbb{R}^3$, $\varphi(x, y, z) = 4x - 5y + 2z$, so $\varphi(1, 2, 3) = 4 - 10 + 6 = 0$.
2. The polynomial functional $\varphi(p) = 3p''(5) + 7p(4)$ is linear by differentiation and evaluation linearity. For $p = x^2$, it gives $3 \cdot 2 + 7 \cdot 16 = 118$.
3. On $\mathcal{P}(\mathbb{R})$, evaluation $p \mapsto p(c)$ and integration $p \mapsto \int_0^1 p(t)\,dt$ are linear by coefficient evaluation and the elementary integral linearity cited in Week 3.
4. If a functional is nonzero, choose v with $\varphi(v) = a \neq 0$. For any $b \in F$, $\varphi((b/a)v) = b$, so the functional is onto.

Definition (Notes Definition 27; LADR 3.112)**.** For an ordered basis $v_1, \dots, v_n$ of V, its dual list is $\varphi_1, \dots, \varphi_n \in V'$ with $\varphi_j(v_k) = \delta_{jk}$, where $\delta_{jk} = 1$ for $j = k$ and 0 otherwise. Explicitly, $\varphi_j(\sum_k a_k v_k) = a_j$. Unique coordinates make this a function; coordinates of a combination add and scale, so it is linear. Any functional with the prescribed basis values must have this formula (Week 3 Proposition 21). We next show this list is a basis, called the dual basis.

Proposition (Notes Proposition 45; LADR 3.111, 3.114, 3.116)

The dual list is a basis of V', so $\dim V' = \dim V = n$. For every $v \in V$ and $\psi \in V'$, $v = \sum_j \varphi_j(v) v_j$ and $\psi = \sum_j \psi(v_j)\varphi_j$.

Where the proof comes from

Want. Obtain expansions in the original and dual bases.

Key step. If $v = \sum a_j v_j$, then $\varphi_k(v) = a_k$ because all other basis values are zero. For functionals, evaluation at v_k gives $\sum_j b_j \varphi_j(v_k) = b_k$, so choosing $b_k = \psi(v_k)$ forces agreement with ψ on the whole basis.

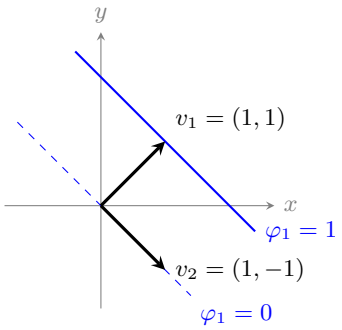
Proof. 1. Write $v = \sum a_j v_j$. Applying φ_k gives $\varphi_k(v) = a_k$, proving the first expansion.
2. If $\sum b_j \varphi_j = 0$, evaluation at v_k gives $b_k = 0$. Thus the dual list is independent.
3. The functional $\sum_j \psi(v_j)\varphi_j$ agrees with ψ at every v_k, so the two linear maps agree everywhere. This gives the second expansion and proves spanning. The dual basis has n vectors, so $\dim V' = n = \dim V$.

□

Example (Notes Example 44; LADR 3.113)
1. In the standard basis of F^n, a functional has row representation $(a_1 \ \cdots \ a_n)$ acting on a coordinate column by $\sum a_j x_j$.
2. The dual basis consists of the coordinate rows with a single 1 in position j, because applying the jth row to e_k gives δ_{jk}.

Compute a nonstandard dual basis

1. In $\mathbb{R}^2$, use $v_1 = (1, 1), v_2 = (1, -1)$. Solve $(x, y) = av_1 + bv_2$ to obtain $a = (x + y)/2$, $b = (x - y)/2$.
2. The coordinate functionals are $\varphi_1(x, y) = (x + y)/2$ and $\varphi_2(x, y) = (x - y)/2$. The four values are $\varphi_1(v_1) = 1$, $\varphi_1(v_2) = 0$, $\varphi_2(v_1) = 0$, and $\varphi_2(v_2) = 1$, by substituting $(1, 1)$ and $(1, -1)$.
3. Equivalently, if Q has columns v_1, v_2 in the standard basis, the rows of Q^{-1} are the dual functionals, because $Q^{-1}Q = I$.



	v_1	v_2
$\varphi_1 = (x + y)/2$	1	0
$\varphi_2 = (x - y)/2$	0	1

The blue lines are sets where φ_1 has a fixed value. This functional reads the v_1 coefficient: it is 1 at v_1 and 0 at v_2. The table gives both dual functionals and all their defining basis values.

Example (Notes Chapter 10: a choice-dependent identification)

Equal finite dimensions give $V \cong V'$, but sending a basis to its dual depends on the chosen basis.

1. For $V = \mathbb{R}$ with basis (1), the dual basis is $\varphi(x) = x$. Sending 1 to φ sends v to the functional $x \mapsto vx$.
2. With basis (2), the dual basis is $\psi(x) = x/2$. Sending 2 to ψ sends $v = (v/2)2$ to $(v/2)\psi$, namely $x \mapsto vx/4$.
3. The two maps differ, for instance at $v = 1$. Thus this construction requires a basis choice.

Shortcuts

- Dual basis functionals return the original basis coefficients, so solving coordinates computes the dual basis.
- To prove an identity of functionals, evaluate both sides on a basis; equality there implies equality everywhere.

Traps

- The dual basis does not generally have the same coordinate rows as the original basis columns: $(1, 1), (1, -1)$ gives rows $(1/2, 1/2), (1/2, -1/2)$.
- The prime in V' does not mean differentiation; a functional such as $p \mapsto p(2)$ returns a scalar, whereas differentiation returns a polynomial.

§4 Double duals and separation

The question. Do all linear functional values distinguish vectors? For $e_1 = (1, 0)$, the functional $(x, y) \mapsto x$ takes value 1, distinguishing e_1 from zero.

Proposition (Basis extension and separation tools for Notes Propositions 46, 50–51)

Every linear functional on a subspace $U \subseteq V$ extends to one on V. If $v \notin U$, there is $\varphi \in V'$ with $\varphi(U) = \{0\}$ and $\varphi(v) = 1$. For arbitrary infinite-dimensional spaces these statements use the basis-extension form of choice cited in Week 2.

Where the proof comes from

Want. Extend a functional and prescribe a value outside U.

Key step. Its values on a basis of U are already fixed. Extend that basis to V and prescribe zero on the added vectors. To separate $v \notin U$, include v among the added basis vectors and prescribe value 1 there; independence is exactly what prevents a conflicting value.

Proof. 1. Choose a basis of U and extend it to a basis of V. Prescribe the given functional values on the first basis and zero on all added basis vectors.
2. For each finite basis expansion, define the new functional by the same coefficient-weighted sum of prescribed values. Unique expansions make it well-defined; adding and scaling expansions proves linearity. Its restriction equals the original functional.
3. If $v \notin U$, adjoining v to a basis of U is independent: a relation with nonzero coefficient on v would put v in U. Extend that list to a basis of V.
4. Prescribe 0 on the basis of U, 1 on v, and 0 on the other basis vectors. The preceding construction gives the separating functional.

□

Definition (Notes Chapter 10: double dual)**.** The double dual is $V'' = (V')'$. Define $j_V(v) : V' \to F$ by $j_V(v)(\varphi) = \varphi(v)$, and $j_V : V \to V''$ by $v \mapsto j_V(v)$.

Proposition (Notes Chapter 10: linearity of evaluation)
For every $v \in V$, $j_V(v) \in V''$, and the map $j_V : V \to V''$ is linear.

Proof. 1. For fixed v, $j_V(v)(a\varphi + \psi) = (a\varphi + \psi)(v) = a\varphi(v) + \psi(v)$. Thus $j_V(v)$ is a linear functional on V'.
2. For every φ, $j_V(av + w)(\varphi) = \varphi(av + w) = a\varphi(v) + \varphi(w) = (aj_V(v) + j_V(w))(\varphi)$. Equality on every functional proves linearity of j_V.

□

Example (Evaluating a double-dual vector)
For $v = (2, -1) \in \mathbb{R}^2$, put $\varphi_1(x, y) = x$, $\varphi_2(x, y) = y$ and $\psi = 3\varphi_1 + 4\varphi_2$.

input to $j_V(v)$	φ_1	φ_2	ψ
value at v	2	-1	$3 \cdot 2 + 4(-1) = 2$

The vector v is fixed; $j_V(v)$ takes a functional as its input and returns that functional's value at v.

Proposition (Notes Proposition 46; LADR §3F Exercise 32(a,c))
The map $j_V : V \to V''$ is injective for every vector space. If V is finite-dimensional, it is an isomorphism.

Proof. 1. For $v \neq 0$, apply separation with $U = \{0\}$ to obtain $\varphi(v) = 1$. Then $j_V(v)(\varphi) = 1$, so $j_V(v) \neq 0$. Thus j_V has zero kernel and is injective.
2. If V is finite-dimensional, $\dim V'' = \dim V' = \dim V$. Week 4 Proposition 31 makes this injective map invertible.
3. Consequently if $\varphi(v) = \varphi(w)$ for all $\varphi \in V'$, then $j_V(v - w) = 0$, and injectivity gives $v = w$.

□

Shortcuts

- To prove $v = 0$, it suffices to prove $\varphi(v) = 0$ for every linear functional, by separation.
- To construct a functional with constraints, choose a basis containing the constrained independent vectors and prescribe its values there.

Traps

- One functional usually cannot distinguish every vector: on $\mathbb{R}^2$, $(x, y) \mapsto x$ cannot distinguish $(0, 0)$ from $(0, 1)$.
- $V \cong V'$ from a basis choice is different from the specified map $j_V : V \to V''$: the former varied in the one-dimensional example, while $j_V(v)(\varphi) = \varphi(v)$ contains no basis choice.

§5 Dual maps, transposes and observables

The question. How does a functional on the output give a functional on the input? For $T(t) = 2t$ and $\varphi(s) = 3s$, the input functional is $(\varphi \circ T)(t) = 6t$.

Definition (Notes Definition 28; LADR 3.118)**.** For $T : V \to W$ linear, the dual map is $T' : W' \to V'$, $T'(\varphi) = \varphi \circ T$. Indeed, $(\varphi \circ T)(av + w) = \varphi(aTv + Tw) = a\varphi(Tv) + \varphi(Tw)$, so the composition is a linear functional. The domain and codomain are reversed.

Proposition (Notes Proposition 47; LADR 3.120)

The map T' is linear. The assignment $\mathcal{D} : L(V, W) \to L(W', V')$, $\mathcal{D}(T) = T'$, is linear, and for $T : U \to V$, $S : V \to W$, $(ST)' = T'S'$.

Where the proof comes from

Want. Compute duals of combinations and compositions.

Key step. The defining value is $(T'\varphi)(v) = \varphi(Tv)$. Linearity of φ handles combinations of maps, and pointwise addition handles combinations of functionals. For a composition, $\varphi(S(Tv))$ first forms $S'\varphi$ and then $T'(S'\varphi)$, explaining the reversed order.

Proof. 1. For $\varphi, \psi \in W'$, $T'(a\varphi + \psi)(v) = (a\varphi + \psi)(Tv) = a\varphi(Tv) + \psi(Tv)$. Thus T' is linear.

2. For $T_1, T_2 : V \to W$, $((aT_1 + T_2)'\varphi)(v) = \varphi(aT_1v + T_2v) = a\varphi(T_1v) + \varphi(T_2v)$. This proves $\mathcal{D}(aT_1 + T_2) = a\mathcal{D}(T_1) + \mathcal{D}(T_2)$.

3. For $\varphi \in W'$, $(ST)'\varphi = \varphi \circ S \circ T = T'(S'\varphi)$, giving the reversed product.

□

Example (Notes Example 45; LADR 3.119) 1. Let $Tp = xp$ on $\mathcal{P}(\mathbb{R})$ and $\varphi(p) = p(2)$. Then $(T'\varphi)(p) = (xp)(2) = 2p(2)$, so $T'\varphi = 2\varphi$.

2. For differentiation $Dp = p'$ and $\psi(p) = p(3)$, $(D'\psi)(p) = p'(3)$.

3. If $\eta(p) = \int_0^1 p(t)\,dt$, then $(D'\eta)(p) = \int_0^1 p'(t)\,dt = p(1) - p(0)$ by termwise integration.

Dualizing the assignment itself (Notes Chapter 10).

1. For $\mathcal{D} : T \mapsto T'$ above, its dual has domain $L(W', V')'$ and codomain $L(V, W)'$ by the definition of a dual map.
2. That definition gives $\mathcal{D}'(\Lambda) = \Lambda \circ \mathcal{D}$, hence $(\mathcal{D}'\Lambda)(T) = \Lambda(T')$ for each T.

Proposition (Notes Proposition 48: double-dual compatibility)

For every linear map $T : V \to W$, $T''j_V = j_W T$, where $T'' = (T')' : V'' \to W''$. This compatibility, together with its basis-free formula, is why j_V is called canonical.

Proof. 1. For $v \in V$ and $\psi \in W'$, $(T''j_V(v))(\psi) = j_V(v)(T'\psi) = (T'\psi)(v) = \psi(Tv)$.

2. The last expression is $j_W(Tv)(\psi)$. Since this holds for every ψ, the two elements of W'' agree, proving the identity.

3. In finite dimension, the j maps are isomorphisms by Proposition 46, so this identity identifies the double-dual map with T through those specified maps.

□

Proposition (Notes Proposition 49; LADR 3.132)

For finite-dimensional V, W, if $A = M_C^B(T)$, then $M_{B'}^{C'}(T') = A^{\mathsf{T}}$, where B', C' are the dual bases. The transpose, defined in Week 3, exchanges rows and columns.

Proof. 1. Write $B = (v_1, \dots, v_n)$, $C = (w_1, \dots, w_m)$, with dual bases φ_k, ψ_j. Then $Tv_k = \sum_i a_{ik} w_i$.
2. $(T'\psi_j)(v_k) = \psi_j(Tv_k) = a_{jk}$. By the functional expansion, $T'\psi_j = \sum_k a_{jk}\varphi_k$.
3. Thus entry (k, j) of the dual matrix is a_{jk}, exactly entry (k, j) of A^{T}.

□

Example (Notes Example 46)

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$ represent $\mathbb{R}^3 \to \mathbb{R}^2$.

1. A functional on $\mathbb{R}^2$ is a row $(x\ y)$. Precomposing with A gives $(x\ y)A = (x + 4y, 2x + 5y, 3x + 6y)$.
2. In dual coordinate columns, the same map is $\begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$, confirming the transpose formula.

Probability convention. A probability column has nonnegative entries summing to 1. A column-stochastic matrix has nonnegative entries and each column sums to 1; entry P_{ij} is the next-state probability of i given present state j. Then $(Pp)_i \geq 0$ and $\sum_i (Pp)_i = \sum_j (\sum_i P_{ij}) p_j = 1$, so Pp is again a probability column. A reward column lists the scalar value assigned to each state; its mean under p is the weighted sum $\sum_i f_i p_i$.

Proposition (Notes Chapter 10: distributions and observables)

For a column-stochastic matrix P, let a probability column p describe the present state and let the real column f list rewards f_i at each state. Their mean reward is $f^{\mathsf{T}} p = \sum_i f_i p_i$. After one step it is $f^{\mathsf{T}} P p = (P^{\mathsf{T}} f)^{\mathsf{T}} p$. The value $(P^{\mathsf{T}} f)_j = \sum_i P_{ij} f_i$ is the conditional mean reward when the present state is j.

Proof. 1. After one step the distribution is Pp, so the mean is $\sum_i f_i \sum_j P_{ij} p_j = \sum_j (\sum_i P_{ij} f_i) p_j$.
2. The inner sum is the mean of the rewards weighted by the conditional probabilities in column j. These inner sums are exactly the entries of $P^{\mathsf{T}} f$, proving the identity.
3. For the row convention, put $A = P^{\mathsf{T}}$ and initial distribution row λ^{T}. For integers $k \geq 0$, with $A^0 = I$ and A^k the repeated product defined in Week 3, $\lambda^{\mathsf{T}} A^k f$ is the mean after k steps, by repeated multiplication and transposition.

□

For matrices of compatible sizes, the dual composition identity gives $(AB)^{\mathsf{T}} = B^{\mathsf{T}} A^{\mathsf{T}}$: the matrix of the dual is the transpose in corresponding dual bases, and $(AB)' = B'A'$. This also follows entrywise because both (i, j) entries are $\sum_k a_{jk} b_{ki}$.

Compute an expected reward in two ways

1. Use Week 3's $P = \begin{pmatrix} 1/2 & 1/4 & 0 \\ 1/2 & 1/2 & 1/2 \\ 0 & 1/4 & 1/2 \end{pmatrix}$, initial $p = e_1$, and rewards $f = (0, 1, 4)^{\mathsf{T}}$. Then $Pp = (1/2, 1/2, 0)^{\mathsf{T}}$ and $f^{\mathsf{T}} P p = 1/2$.
2. Alternatively, its entries are $(1/2)0 + (1/2)1 = 1/2$, $(1/4)0 + (1/2)1 + (1/4)4 = 3/2$, and $(1/2)1 + (1/2)4 = 5/2$. Thus $P^{\mathsf{T}} f = (1/2, 3/2, 5/2)^{\mathsf{T}}$; its first entry gives the same mean from state 1.

Shortcuts

- Compute a dual map by precomposition; its matrix in dual bases is the transpose, so no new matrix solving is needed.
- For expected rewards, choose whichever is shorter: update p by P or update f by P^{T}.

Traps

- Dual maps reverse direction: a 2×3 matrix gives a dual matrix of size 3×2, not 2×3.
- Transposition does not reverse entries within a row; it exchanges indices. Example 46 explicitly gives the correct array.
- Unreversed dual products may be undefined: for $T : \mathbb{R} \to \mathbb{R}^2$ and $S : \mathbb{R}^2 \to \mathbb{R}^3$, $T'S'$ is defined, but $S'T'$ has mismatched intermediate spaces.

§6 Annihilators and rank of the dual map

The question. Which functionals vanish on a specified subspace? For the plane spanned by $(1,1,0), (0,1,1)$, the row $(-1,1,-1)$ vanishes on both generators, hence on the whole plane.

Definition (Notes Definition 29; LADR 3.121)**.** The annihilator of a subspace $U \subseteq V$ is $U^0 = \{\varphi \in V' : \varphi(u) = 0 \text{ for every } u \in U\}$. The same definition makes sense for any subset U (the textbook convention), and the subspace proof below still applies. The ambient space V determines which functionals are allowed. Hereafter U is a subspace unless specified otherwise.

Example (LADR 3.122–3.123) 1. Let $U = x^2\mathcal{P}(\mathbb{R}) \subseteq \mathcal{P}(\mathbb{R})$ and $\varphi(p) = p'(0)$. If $p = x^2 q$, then $p'(0) = (2xq + x^2 q')(0) = 0$, so $\varphi \in U^0$. It is a nonzero functional, since $\varphi(x) = 1$.

2. In $\mathbb{R}^5$, let $U = \operatorname{span}(e_1, e_2)$ and let φ_j be the standard coordinate functionals. A row $(a_1, a_2, a_3, a_4, a_5)$ vanishes on e_1, e_2 exactly when $a_1 = a_2 = 0$. Thus

$$U^0 = \operatorname{span}(\varphi_3, \varphi_4, \varphi_5), \qquad \psi = \underbrace{0\varphi_1 + 0\varphi_2}_{\text{forced by } \psi(U)=0} + \underbrace{a_3\varphi_3 + a_4\varphi_4 + a_5\varphi_5}_{\text{arbitrary coefficients}}.$$

The three remaining coordinate functionals are independent by evaluating at e_3, e_4, e_5.

Proposition (LADR 3.124–3.125, 3.127)

U^0 is a subspace of V'. In finite dimension, $\dim U^0 = \dim V - \dim U$. The extreme-case conclusions extend LADR 3.127 to all dimensions using separation: $U^0 = \{0\}$ iff $U = V$, and $U^0 = V'$ iff $U = \{0\}$.

Proof. 1. The zero functional annihilates U, and if φ, ψ do, then $(a\varphi + \psi)(u) = 0$ for every $u \in U$. Thus U^0 is a subspace.

2. In finite dimension, extend a basis $u_1, \ldots, u_k$ of U to $u_1, \ldots, u_n$ in V and take its dual basis $\varphi_1, \ldots, \varphi_n$. A functional annihilates U iff its first k coefficients in this dual basis are zero. Thus $\varphi_{k+1}, \ldots, \varphi_n$ is a basis of U^0, of length $n - k$.

3. If $U = V$, only the zero functional vanishes everywhere. If $U \neq V$, separation supplies a nonzero functional annihilating U, so $U^0 \neq \{0\}$.

4. If $U = \{0\}$, every functional vanishes there. If U contains a nonzero u, separation supplies a functional with value 1 at u, so $U^0 \neq V'$.

□

Example (Notes Example 47)

Let $U = \operatorname{span}((1,1,0), (0,1,1)) \subseteq \mathbb{Q}^3$.

1. A row (a, b, c) annihilates both generators iff $a + b = 0$ and $b + c = 0$.
2. Thus $(a, b, c) = b(-1, 1, -1)$, so $U^0 = \operatorname{span}((-1, 1, -1))$.
3. The two generators are independent by their first and last coordinates, so $\dim U = 2$ and $\dim U^0 = 1$, verifying $2 + 1 = 3$.

Proposition (Notes Proposition 50; LADR 3.128(a), 3.130(b))

For any linear $T : V \to W$, $\ker T' = (\operatorname{range} T)^0$ and $\operatorname{range} T' = (\ker T)^0$. The general-dimensional statement uses the functional-extension tool above.

Where the proof comes from

Want. Express the kernel and range of T' using T.

Key step. $T'\psi = 0$ means $\psi(Tv) = 0$ for every v, exactly vanishing on $\operatorname{range} T$. To obtain $\varphi = T'\eta$, the required values must be $\eta(Tv) = \varphi(v)$. If $Tv = Tw$, then $v - w \in \ker T$, so a functional φ vanishing there assigns the same value. Define η on $\operatorname{range} T$ this way and extend it to W.

Proof. 1. $\psi \in \ker T'$ iff $\psi(Tv) = 0$ for all v, exactly $\psi \in (\operatorname{range} T)^0$.

2. If $\varphi = T'\psi$ and $v \in \ker T$, then $\varphi(v) = \psi(Tv) = \psi(0) = 0$. Thus $\operatorname{range} T' \subseteq (\ker T)^0$.

3. For $\varphi \in (\ker T)^0$, define $\eta : \operatorname{range} T \to F$ by $\eta(Tv) = \varphi(v)$. If $Tv = Tw$, then $v - w \in \ker T$, so $\varphi(v) = \varphi(w)$; the rule is well-defined.

4. It is linear: $\eta(aTv + Tw) = \eta(T(av + w)) = \varphi(av + w) = a\varphi(v) + \varphi(w)$. Extend η to $\widetilde{\eta} \in W'$ by the extension tool. Then $T'\widetilde{\eta}(v) = \widetilde{\eta}(Tv) = \varphi(v)$, proving the reverse inclusion.

□

Proposition (Notes Proposition 51; LADR 3.129, 3.131)

For a linear map $T : V \to W$, T is injective iff T' is surjective; T is surjective iff T' is injective.

Proof. 1. T injective $\iff \ker T = \{0\} \iff (\ker T)^0 = V' \iff \operatorname{range} T' = V' \iff T'$ surjective.

2. T surjective $\iff \operatorname{range} T = W \iff (\operatorname{range} T)^0 = \{0\} \iff \ker T' = \{0\} \iff T'$ injective.

□

Proposition (LADR 3.128(b), 3.130(a), 3.133)

For a linear map $T : V \to W$ between finite-dimensional spaces, $\operatorname{rank} T' = \operatorname{rank} T$ and $\dim \ker T' = \dim \ker T + \dim W - \dim V$. Consequently row rank equals column rank for every finite matrix.

Proof. 1. $\dim \operatorname{range} T' = \dim(\ker T)^0 = \dim V - \dim \ker T = \operatorname{rank} T$.

2. $\dim \ker T' = \dim(\operatorname{range} T)^0 = \dim W - \operatorname{rank} T = \dim W - \dim V + \dim \ker T$.

3. For a matrix A, its column rank is the rank of its map by Week 3. The dual matrix is A^{T}, whose column rank is the row rank of A. Equality of map ranks gives the equality of matrix ranks, confirming Week 3's factorization proof.

□

Shortcuts

- To compute an annihilator, write a general row and set its value on each subspace basis vector equal to zero.
- Finite-dimensional annihilator dimensions supply a quick check: subspace dimension plus annihilator dimension equals ambient dimension.

Traps

- Annihilator containment reverses subspace containment: more vectors to vanish on means fewer functionals. For $0 \subset \mathbb{R}e_1 \subset \mathbb{R}^2$, the annihilators are all rows, rows $(0, b)$, and zero.
- Equal rank of T and T' does not give equal nullity when domain and codomain dimensions differ: the map $\mathbb{R}^2 \to \mathbb{R}$, $(x, y) \mapsto x$, has nullity 1, but its dual has nullity 0.

§7 Problem set

HW5 P1. Let $T : \mathcal{P}_3(\mathbb{R}) \to \mathbb{R}^2$, $Tp = (p(0), p(1))$. (a) Determine $\ker T$ and prove surjectivity. (b) Describe the induced isomorphism $\mathcal{P}_3(\mathbb{R})/\ker T \to \mathbb{R}^2$. (c) Find a quotient basis. *Additional asks from the same map in Practice Midterm 2, Question 1:* verify linearity and specify the images of the quotient-basis elements.
Hint: Factor polynomials vanishing at 0 and 1; use linear polynomials for arbitrary outputs.

HW5 P2. If $T : V \to W$ is linear and U is a subspace of V with $U \subseteq \ker T$, prove there is a unique linear $\widetilde{T} : V/U \to W$ with $\widetilde{T}(v + U) = Tv$. Determine its kernel.
Hint: Use Proposition 42 with target subspace zero, then test the zero output.

HW5 P3 (LADR 3F.1). Prove every linear functional $\varphi : V \to F$ is either zero or surjective.
Hint: Scale a vector where the value is nonzero.

HW5 P4. For a basis $v_1, \dots, v_n$ and its dual basis $\varphi_1, \dots, \varphi_n$, prove (a) $v = \sum_j \varphi_j(v) v_j$ for each $v \in V$ and (b) $\psi = \sum_j \psi(v_j)\varphi_j$ for each $\psi \in V'$.
Hint: Apply a dual functional to a basis expansion, then evaluate the second identity on each v_k.

HW5 P5. Let $T : V \to W$ be linear between finite-dimensional spaces and let A represent it in bases B, C. Prove $T' : W' \to V'$ has matrix A^{T} in the corresponding dual bases.
Hint: Compute $(T'\psi_i)(v_j)$, as in Proposition 49.

HW5 P6. Let $U = \operatorname{span}((1,1,0),(0,1,1)) \subseteq \mathbb{R}^3$. (a) Find U^0, identifying functionals with rows. (b) Verify $\dim U + \dim U^0 = 3$.
Hint: A general row must vanish on both generators.

HW5 P7 (optional). For $j_V(v)(\varphi) = \varphi(v)$, prove $j_V : V \to V''$ is injective, and is an isomorphism when V is finite-dimensional.
Hint: Use separation to detect a nonzero vector and the equality of dual dimensions.

Exam practice (practice-midterm2.pdf, Question 2). Let V, W, Z be finite-dimensional real vector spaces, and let $T : V \to W$, $S : W \to Z$ be linear. (a) Define the dual map $T^* : W^* \to V^*$ and prove it is linear. (b) Prove $(S \circ T)^* = T^* \circ S^*$. (c) Prove that if T is surjective, then T^* is injective. (d) For $V = \mathbb{R}^2$, $W = \mathbb{R}$, $T(x, y) = 2x - y$, and $\ell(t) = 3t$, compute $T^*\ell$. In this exam, $*$ denotes the same dual-space and dual-map notation as our prime.
Hint: Use precomposition for (a,b), and evaluate a functional on all outputs of T for (c).

Exam review (Midterm 2 Review, Problem 1). Let $V = \operatorname{span}(1, \sin x, \sin 2x)$ on $[-\pi, \pi]$. Define $\delta_0(f) = f(0)$, $\delta_{\pi/2}(f) = f(\pi/2)$ and $\delta_0'(f) = f'(0)$. (a) Prove these three functionals form a basis of V'. (b) Find the dual basis $\varphi_1, \varphi_2, \varphi_3$ to $(1, \sin x, \sin 2x)$ as combinations of those functionals. (c) Express $L(f) = \pi^{-1} \int_0^\pi f(x)\,dx$ in both functional bases.
Hint: For $f = a + b \sin x + c \sin 2x$, compute the three values and solve for a, b, c.

Exam review (Midterm 2 Review, Problem 2). Let finite-dimensional V contain a subspace U. Let $i : U \to V$ be inclusion and $\pi : V \to V/U$ the quotient map. (a) Find $\ker i'$, $\operatorname{range} i'$, $\ker \pi'$ and $\operatorname{range} \pi'$ as subspaces/annihilators. (b) Construct an isomorphism $U' \to V'/\operatorname{range} \pi'$.
Hint: The dual of inclusion restricts a functional to U; extend a functional on U and take the coset of that extension.

Exam review (Midterm 2 Review, Problem 8, dual-map assertion). True or false: if T is injective, then T' is injective. Prove it or give a counterexample.
Hint: Consider $T : \mathbb{R} \to \mathbb{R}^2$, $T(t) = (t, 0)$, and write out precomposition of a general row.